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Distributed Machine Learning Training in Oil and Gas Industry via Federated Learning

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Machine Learning (ML) solutions are becoming a cornerstone of reliable data interpretation and various task automation in Oil and Gas (O&G) exploration and production. The conventional approach in ML training requires centralizing the entire training dataset, which presents challenges like data privacy, residency, scalability, heterogeneity, and version control. To alleviate some of these challenges, we propose a Federated Learning (FL) approach for distributed ML model training in the industry, particularly in conveyance applications, enabling collaborative learning across geographically diverse datasets and ensuring adaptability to varied sources.

Typically, the first step in ML model training is to obtain and centralize data to enable model training on specific computers. In protected industries, data often must reside locally to ensure privacy. In such cases, where data residency is paramount, one can deploy distributed training through FL. In FL approach, ML models are trained locally on diverse computers on their local, domain-specific data and they only periodically communicate their model parameters with the central server for model aggregation. This approach produces a consolidated and aggregated central model that never sees the full data but leverages data from all the sources for a globally generalized ML model.

We implemented FL deployment on geographically diverse computers where central server initializes and distributes global model parameters to each of the client computers where the private data resides, and each local client computer trains only on their datasets. After local training, the parameters of each local model are transmitted to central server for aggregation over the internet through Virtual Private Network (VPN) access and aggregated global model is redistributed back to the clients for further training if needed. The Pythonic FL framework makes the training process model-agnostic, allowing a variety of ML models to be trained with minimal changes. We demonstrate two successful examples in the context of wireline conveyance automation: one involving the training of an XGBoost classifier to predict high-tension events in pre-job scenario using specific job parameters such as cable properties, well and fluid information, tool specifications etc. and the second focuses on cable tension prediction using multi-layer perception (MLP) as a regression problem.

Introduction

ML model training benefits significantly from diverse datasets to avoid overfitting while it allows model generalization for global deployment. The current approach in O&G industry is to obtain required permissions from each data owner to centralize collected data and ensure its safety until it is purged after training. While this approach provides benefits to the trained model, it is crucial that all required steps be followed to ensure data safety over the course of the training, validation and deployment. Centralized ML training also brings all the computational resources together to the central location where the model training and validation is carried out.

At the start of this work, we obtained the required permissions from data owners, collected multiple field datasets from various field engineers, anonymized the data and centralized the training process to ensure the problems that we aim to solve are feasible given the collected features. After training centralized ML models, we repeated the training in distributed settings via FL across several computers spanning the USA. To this extend, we demonstrate both classification and regression tasks can be adapted where the data resides in different states in the continental United States connected via VPN over the internet. In the FL training scenario, the local models are trained on local computers and model parameters are aggregated at the central server, which never has access to the datasets.

The first problem we tackled was to train a classifier model that can utilize pre-job settings of a wireline job to predict if there would be any high-tension event such as stuck tool, key-seating or any type of out of ordinary high tension observed. A binary label is assigned to each historical job that we collected based on the reported sequence of events and measured cable tensions, where label one indicates high-tension event observed, and label zero indicates normal operation.

In the second problem we tackled, we trained MLP models in regression tasks, where the neural network would receive wireline job parameters, as well as cable velocity profile to be followed during the job to predict the head and surface tensions of the conveyance cable. Head-tension predictions are the tension measured at the head of the tool, where cable connects to the borehole assembly (BHA) and surface-tension predictions are the tension measured at the surface. We initially centralized the training to ensure the problem is solvable with the collected features with an MLP model. Once this ground truth was established and performance benchmarked, we deployed the FL training framework on several geographically diverse local computers and trained global models to compare the results. Training via FL not only preserves data privacy in training global models but also enables the personalization of the global model to local needs if further data is collected, streamlines basin-level model development, and enhances adaptability to regional variations in O&G operations.

Introduction to Federated Learning

Federated Learning (FL) is a distributed machine learning training technique that allows multiple devices to train a shared model on their local datasets without directly centralizing the data or the computational resources. These devices, i.e. clients, might belong to different organizations, entities, or individuals for whom private data belongs, geographically dispersed and can be used to train the global model. FL addresses certain challenges that centralized ML training faces but also brings in new ones that the user may not be aware of. Primary benefits are the privacy and security of sensitive private data, which always stays on the local devices during training where only the computational resources belonging to the data owner operate on it. This reduces the risk of any data breach if data were centralized, helps with improved compliance regulations by keeping the data localized, and allows access to diverse private datasets that may be otherwise inaccessible due to local laws and regulations.

On the other hand, the primary challenges are introduced by system complexity and heterogeneity to communicate with and train on geographically diverse clients, communication overhead due to increased number of computational resources residing different locations, data heterogeneity based on the client

data collection schema and storage principles, as well as sophisticated attacks to recover the model parameters during communication to access localized private data. Some of these challenges are addressed algorithmically such as introduction of differential privacy, secure aggregation protocol as well as homomorphic encryption [1]. Others are addressed through securing the physical layers such as deployment of secure network infrastructures, physical security of client devices, and secure communication channels like VPNs. These topics are not in the scope of this work. Another benefit of FL approach is to maximize utilization of heterogeneous distributed compute resources as well as limited datasets [2]. It has been demonstrated in distributed large- language model (LLM) training can be tackled various computational sources via adaptable FL approach that can achieve or surpass performance of centrally trained homogenous resources. This topic is also not in the scope of this work.

Binary Classification Task of High-Tension Events Classification

One of the challenges in wireline conveyance is to predict the possibility of a high-tension event happening during a job based on the pre-job design parameters set by the field engineers for the specific task at hand. High-tension events could be caused by key-seating of the cable, differential sticking of the tool or cable, tool getting stuck for certain known reason such as stick-and-slip or completely unknown reasons. In any of the cases, high-tension events are disruptive and should be avoided for safe and speedy operation.

The premise of this binary classification task is that historical data on the various wireline jobs can tell an indication of which pre-job settings can potentially suggest high tensions event could happen. For this task, approximately 100 diverse datasets were collected with proper permissions, anonymized and centralized for centralized ML training. Nearly 20 features were collected for each of the jobs, which includes tool string parameters such as length, weight and volume; cable properties such as diameter, volume and densities in air and water; borehole fluid and mud parameters such as density and weights. Also, borehole parameters such as the cased and open-hole radius, dog-leg severity and inclination are useful parameters. Utilizing these pre-job parameters we also created several new features for the ML model through nonlinear combination.

XGBoost (eXtreme Gradient Boosting) algorithm is an open-source ML method that has become the de facto ML solution for tabular datasets and is preferred choice for many scientists and ML engineers. A brief study on our centralized dataset confirmed the high performance of XGBoost hence selected here for further study in FL setting. XGBoost is an ensemble learning method using decision trees as the weak learners and gradient boosting to train weak learners sequentially.

The dataset was split into 80% training and 20% testing and a 5-fold cross validation schema was deployed to study the hyperparameters via grid search method. The list of hyperparameters that were studied are shown in Table 1 with their ranges. Best parameter combination of the hyperparameter using this centralized set was {'colsample_bytree': 0.9, 'learning_rate': 0.2, 'max_depth': 5, 'n_estimators': 10, 'subsample': 0.9} with cross-validation accuracy of 0.70.

Table 1—List of hyperparameters search on the centralized dataset via grid search, ranges tested and description of the parameters.

Hyperparameter	Values	Description
n_estimators	[10, 20, 50, 100]	Number of boosting rounds
learning_rate	[0.01, 0.05, 0.1, 0.2]	Step size shrinkage
max_depth	[5, 7, 10, 15]	Maximum depth of a tree
subsample	[0.7, 0.8, 0.9]	Subsample ratio of the training instance
colsample_bytree	[0.7, 0.8, 0.9]	Subsample ratio of columns when constructing each tree

Best model using these hyperparameters achieved an accuracy of 80% on the test set. Confusion matrix for this model's prediction of test set is shown below in [Figure 1](#).

	0	1
0	10	2
1	2	6

Figure 1—Confusion matrix of the best model prediction on the test set after hyperparameter grid search using the centralized data.

In the FL implementation of the XGBoost binary classifier training, the centralized data was divided into three separate sets and sent to three separate clients in locations spanning from Massachusetts to California. These clients were tasked with training their local models, while a central server that never sees any of the data but coordinates the FL training, communication and aggregation of the central model. One should note that the datasets are significantly reduced to approximately 33 jobs, much less compared to full set in centralized training scenario, which can lead to biased local models and drop in the aggregated model performance can be expected. Flower Federated Learning framework is utilized in this work [1], which is platform agnostic implementation of FL allowing ML model training in well-known frameworks like scikit-learn, TensorFlow or PyTorch. For server strategy, built-in "FedXgbBagging" is used. Utilizing the best hyperparameters from the above study, FL launched with evaluation metric of "AUC" and all three clients are used for fitting and evaluation. Clients ran for 15 local epochs to train this binary classification task for 5 iterations of the federated learning. Since XGBoost works efficiently on small datasets, training was only carried on central processing units (CPUs) of the client computers.

A high AUC score of 0.9 was achieved after few trials, indicating a very good, aggregated model as seen in [Fig 2.b](#). To test the trained aggregated XGBoost classifier, whole dataset was used utilized and the accuracy, weighted-average precision, recall and F1 scores are computed as seen in [Fig 2.c](#). For all metrics, over 80% is observed indicating a successful aggregated model trained on distributed dataset via FL comparable to the centrally trained model performance.

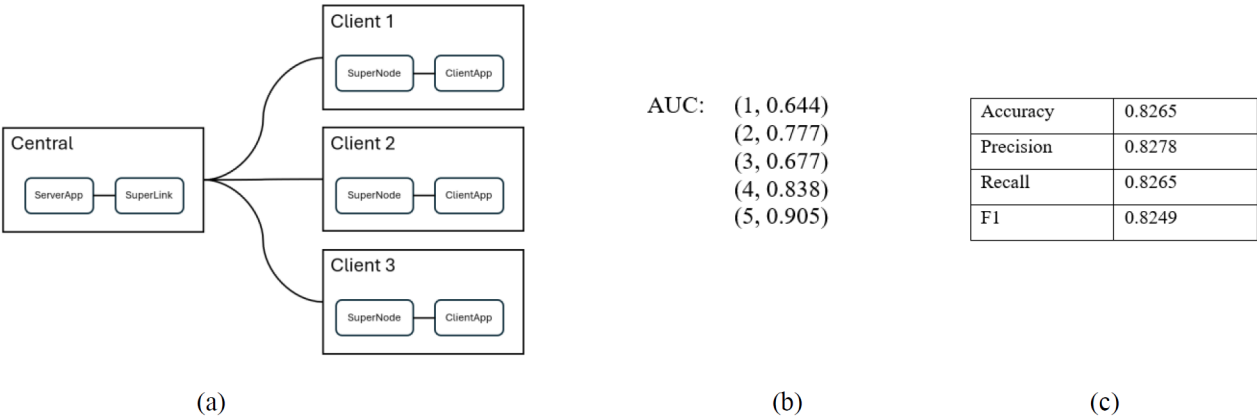


Figure 2—(a) FL setup of Central computer with three clients spread over the continental USA. Central computer initiates SuperLink on server side, while each client initiates SuperNodes on their counterparts. Such connection allows the ServerApp and ClientApps to communicate [3], (b) Aggregated AUC score over five iterations of the FL training, (c) Accuracy, Precision, Recall and F1 score of the aggregated model on full dataset for comparison purposes.

Regression Task of Cable Tension Prediction

Cable tension measurements at the head of the tool assembly and at the surface are carried out due to safety and efficiency reasons. To carry a safe conveyance job at an optimum speed, operators and wireless job design engineers need to be able to accurately predict these tensions so that proper safety mechanisms can be put in place along with the most appropriate tool selection for the job. In addition to safety concerns, speedy operations can reduce both cost and rig side emissions. Through accurate prediction of these tension, one can accurately estimate the depth, prevent wireline sticking events, or avoid damage to tools or borehole itself. Prediction of these tension measurements is a complicated task that depends on the selected tools, accessories, borehole geometry, fluids deployed, and various other criteria. Former attempts to model cable tensions and borehole behavior with physics-based models have achieved limited success as some complex and dynamic behavior cannot be captured fully or might be unique to the geology of the basin.

An alternative to physics-based models is to train ML model that utilizes extensive set of archived conveyance jobs to learn how to predict the head-tension and surface-tension for potential future jobs. Given enough diverse conveyance jobs that contains various features affecting the cable tension behavior, ML models can be trained without needing to completely understand all potentially possible complex physical phenomena in the borehole.

Multi-layer Perceptron (MLP), also known as feed-neural networks (NN) are trained in this work using several conveyance job parameters as inputs to predict the surface tension and head tension. Approximately 100 field datasets from various wells in a basin were collected, anonymized and unified for the training centrally. Each well had multiple jobs with varying tool settings to carry out different wireline measurements. Approximately 4-5 jobs per well corresponds to approximately 25 different wells that we had access through collected data. Centralized data was split into approximately 50% training and 50% testing, where at least 2 jobs per well was in the training to ensure ML model was exposed to all wells. Data was extensively studied to ensure the noisy or failed jobs were kept in the test set to prevent outliers from skewing the training. Over 30 million total data points were collected for training of wireline conveyance jobs to predict the surface and head tension in run-in-hole (RIH) and pull-out-of-hole (POOH) scenarios. Input variables include cable and tool properties, well dimensions, inclination, azimuth and curvature, fluid density, cased/open-hole indicator as well as the cable speed at depth. Due to NN sensitivity to scales of the input variables, normalization and standardization was also carried out. In the centralized training scenario, a simple two-layer MLP with 128 neurons was sufficient to predict these two outputs in a regression task setting as shown on a test job in [Fig.3.a](#). Surface-tension measurements and MLP predictions are shown on the top section, while the head-tension measurements and predictions at the bottom section over time.

For the FL training implementation, same dataset was split into three datasets, where division respected the well locations. Each client computer received approximately 8 wells with several jobs per well for training and few jobs for validation. None of the clients were aware of the datasets in other clients to ensure the data is private while the local models are trained. Utilizing the same MLP architecture with two layers and 128 neurons with ReLU non-linear activation function, FL training was conducted on three client computers spread from Massachusetts to California within the continental USA. Built-in "FedAvg" aggregation was used to aggregate models, where local training was run for one epoch for nearly 10 million datapoints per client. Total of three federated learning iterations were sufficient to achieve a comparable model. A test case performance of the aggregate model is shown in [Fig.3.b](#). where none of the local models encountered this dataset during training. FL trained model performance is comparable to the centrally trained ML model.

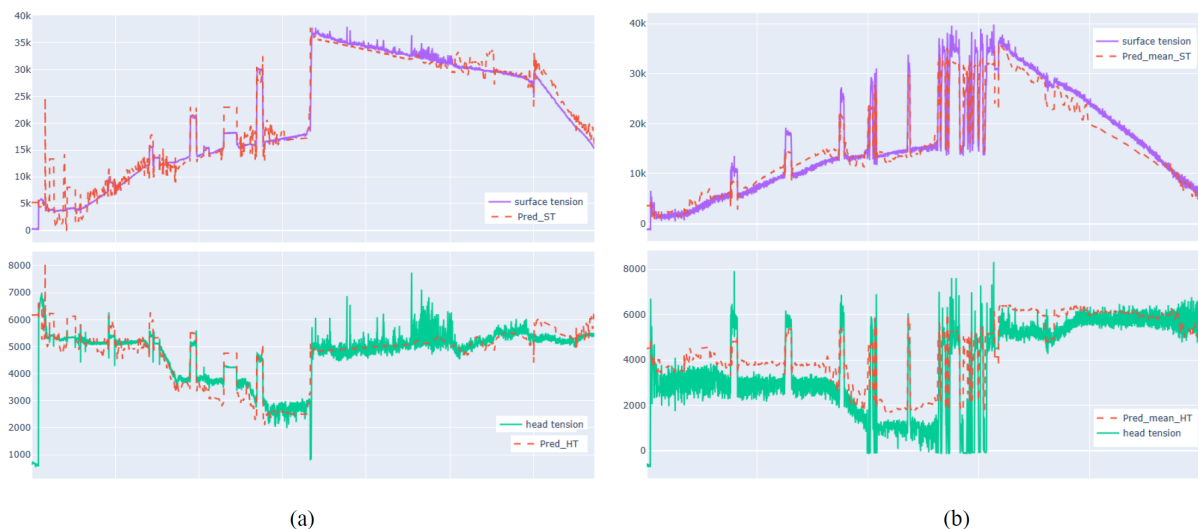


Figure 3—(a) Centrally trained MLP model prediction on a new well that was not in the training set, (b) Federated Learning trained aggregated MLP model prediction on a new well that was not in any of the client datasets. Vertical axis is in units of kN, horizontal axis is the time.

Discussion

This work demonstrates that Federated Learning can be utilized to train ML models in various scenarios for classification or regression tasks in wireline conveyance jobs. The FL framework is language agnostic but requires the ML engineer to understand how it operates to communicate as well as how to set up the problem over geographically distributed computational resources. While it alleviates the problem with data residency, client computers are needed to be free to run FL training which might take away the resources from the field deployed machines. Because this work was carried out on a secure network, minimal security protocols were activated to reduce the communication burden. If the training were to be carried out over unsecured networks, additional protocols must be turned on causing extra communication overhead to ensure the shared model parameters remain unchanged and uncompromised during transit.

Similar challenges to centrally trained ML approaches still exist in the FL training when it comes to understanding the features of the problem, cleaning the datasets, unifying data variables for training and choosing the right hyperparameters to train the best models. In fact, some of these problems may have been exacerbated as the ML engineer may not have direct access to certain private data and not know the outliers or certain distributions that may exist in each separate client's repository. Because the model parameters are aggregated, one can expect that some local models may perform better while others may drag the performance down. This could possibly be due to lack of diversity in certain client datasets or unexplained noise that challenges the training. Training times were comparable given the fact that we exploited the prior knowledge on the problem, on the dataset variables and on choosing the best hyperparameter combinations for best performance. We can recommend conducting initial small-scale study of variables, the algorithm selection, and hyperparameter tuning before launching the full-scale FL training on multiple clients.

References

1. McMahan, H. B., Moore, E., Ramage D., Hampson, S., Agüera y Arcas, B., 2017. Communication-Efficient Learning of Deep Networks from Decentralized Data. Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS), Ft. Lauderdale, FL, USA, April 20 - 22, 2017. <https://doi.org/10.48550/arXiv.1602.05629>
2. Lorenzo, S., Iacob, A., Cao, Z., Marino, B., Gao, Y., Paulik, T., Zhao, W., Shen, W. F., et al, 2024. The Future of Large Language Model Pre-training is Federated. Proceedings of the International

Workshop on Federated Foundation Models In Conjunction with NeurIPS 2024, Vancouver, BC, Canada, Dec 15, 2024. <https://doi.org/10.48550/arXiv.2405.10853>

3. *Flower A Friendly Federated AI Framework*: <https://flower.ai/>